

Moduli Reading Group, Lect 4 Notes (Very Informal!)
 Pijush, 31st July 2026.

Definition 3.1 An algebraic group over k is:

- A scheme G over k
- with morphisms:
 - $m: G \times G \rightarrow G$ (group law)
 - $e: \text{Spec } k \rightarrow G$ (identity)
 - $i: G \rightarrow G$ (inverse)

such that we have the following commutative diagrams:

(i) Associativity:

$$\begin{array}{ccc}
 G \times G \times G & \xrightarrow{\text{id} \times m} & G \times G \\
 m \times \text{id} \downarrow & \curvearrowright & \downarrow m \\
 G \times G & \xrightarrow{m} & G
 \end{array}$$

(ii) Mult by identity:

$$\begin{array}{ccccc}
 \text{Spec } k \times G & \xrightarrow{e \times \text{id}} & G \times G & \xleftarrow{\text{id} \times e} & G \times \text{Spec } k \\
 & \searrow \cong & \downarrow m & \swarrow \cong & \\
 & & G & &
 \end{array}$$

$$\begin{array}{ccccc}
 \text{(iii)} & G & \xrightarrow{(\text{id}, i)} & G \times G & \xleftarrow{(i, \text{id})} G \\
 & \downarrow & & \downarrow m & \downarrow \\
 & \text{Spec } k & \xrightarrow{e} & G & \xleftarrow{e} \text{Spec } k
 \end{array}$$

(Mult by inverse)

Remark: G is assumed to be locally of finite type over k .

Affine algebraic group.

Homomorphism of algebraic groups: Morphism $f: G \rightarrow H$ of schemes st. f preserves group law:

$$\begin{array}{ccc}
 G \times G & \xrightarrow{m_G} & G \\
 f \times f \downarrow & \curvearrowright & \downarrow f \\
 H \times H & \xrightarrow{m_H} & H
 \end{array}$$

$\mathcal{O}(G) := k$ -algebra of regular functions on G .

Examples: ① $G_a := \text{Spec } k[t]$. (Called the additive group).

Underlying scheme: A^1 .

Mult $m: G \times G \rightarrow G$ is given by $m^*: k[t] \rightarrow k[t] \otimes k[t]$
 $t \mapsto t \otimes 1 + 1 \otimes t$

$e: \text{Spec } k \rightarrow G$ is given by $\left(\begin{array}{l} \text{or } k[t] \otimes k[t] \cong k[z, w] \\ \text{then } t \mapsto z + w \end{array} \right)$
 $e^*: k[t] \rightarrow k$
 $t \mapsto 0$

$i: G \rightarrow G$, $i^*(t) = -t$.

Note: The functor of points $h_{G_a} := \text{Hom}(-, G_a)$.

Then $G_a: k\text{-schemes} \rightarrow \text{Groups}$

$$R \longmapsto G_a(R) := \text{Hom}_{k\text{-alg}}(\text{Spec } R, G_a) \\ = \text{Hom}_{k\text{-alg}}(k[t], R) \cong (R, +).$$

② $G_m := \text{Spec } k[t, t^{-1}]$ (Mult group).

Underlying scheme: $A^1 \setminus \{0\}$

Mult map: $m: G_m \times G_m \rightarrow G_m$ given by

$$m^*: k[t, t^{-1}] \rightarrow k[t, t^{-1}] \otimes k[t, t^{-1}] \\ t \mapsto t \otimes t$$

$$e^*: k[t, t^{-1}] \rightarrow k \\ t \mapsto 1$$

$$i^*: k[t, t^{-1}] \rightarrow k[t, t^{-1}] \\ t \mapsto t^{-1}$$

$$\left(\begin{array}{l} \text{or } k[t, t^{-1}] \otimes k[t, t^{-1}] \cong k[z, w]_{(z, w)} \\ t \mapsto z \otimes w \end{array} \right)$$

③ $GL_n := \text{Spec} \left(k[x_{ij} \mid 1 \leq i, j \leq n]_{\det(x_{ij})} \right)$

Closed points: Invertible $n \times n$ matrices.

$$m^*(x_{ij}) = \sum_{k=1}^n x_{ik} \otimes x_{kj}$$

$$e^*(x_{ij}) = \delta_{ij}$$

$$i^*(x_{ij}) = (x_{ij})^{-1}_{ij}$$

where $(x_{ij})^{-1}_{ij}$ is the regular fn that takes ij entry of inverse of a matrix.

$$\textcircled{7} \quad GL(V) := \text{Spec} \left(k\text{-algebra of invertible linear transforms} \right) \\
V \rightarrow V \text{ where mult} = \text{composition} \\
\cong GL_{\dim(V)}.$$

$$GL(V)(R) = GL_{\dim(V)}(R) = \text{dim } V \times \text{dim } V \text{ invertible matrices in } R \\
= \text{automorphisms } R \otimes_k V \rightarrow R \otimes_k V \\
= \text{Aut}(R \otimes_k V).$$

• Action of an affine alg grp G on a scheme X :

- A morphism $\sigma: G \times X \rightarrow X$ s.t.

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{(m, \text{id})} & G \times X \\ \downarrow (\text{id}, \epsilon) & \nearrow & \downarrow \sigma \\ G \times X & \xrightarrow{\epsilon} & X \end{array} \qquad \begin{array}{ccc} \text{Spec } k \times X & \xrightarrow{(e, \text{id})} & G \times X \\ & \searrow \cong & \downarrow \sigma \\ & & X \end{array}$$

- G -equivariant morphism $f: X \rightarrow Y$ of alg groups with actions σ_X and σ_Y resp.

- G -invariant.

↖ (skip proof).

• Theorem 3.9: If G is affine alg group, then $G \leq GL_n$.

Proof outline: - Consider $\mathcal{O}(G)$ and its finite generators as k -alg. $f_1, \dots, f_n$

- Let $W := k\langle f_1, \dots, f_n \rangle$. By Lemma, $W \subseteq V \subseteq \mathcal{O}(X)$ s.t. $\dim(V) < \infty$ and V is G -inv.

- Consider $m^*: \mathcal{O}(G) \rightarrow \mathcal{O}(G) \otimes \mathcal{O}(G)$

By G -invariantness of V , $m^*(f_i) \in \mathcal{O}(G) \otimes V$ as regular function.

Write $m^*(f_i) = \sum_{j=1}^n a_{ij} \otimes f_j$

- This gives hom $G \rightarrow \text{Aut}(\mathcal{O}(X))$
 $g \mapsto \sum a_{ij}(g) f_i$

This gives hom $\rho^*: \mathcal{O}(M_n) \rightarrow \mathcal{O}(G)$
 $x_{ij} \mapsto a_{ij}$

- This ρ^* is surj, so $\rho: G \rightarrow M_n$ is a closed embedding

- Further, ρ is a homomorphism of semi-groups with mult on M_n

- Since G is a group, therefore, $\rho(G)$ will be a subgroup inside GL_n .

Defⁿ 3.10: Let G be an affine algebraic group scheme over k .

(1) G is an **algebraic torus** if $G \cong \mathbb{G}_m^n$ for some $n > 0$.

(4) **Character Group** of torus T is $X^*(T) := \text{Hom}(T, \mathbb{G}_m)$

(5) **Cocharacter Group** of torus T is $X_*(T) := \text{Hom}(\mathbb{G}_m, T)$
 (Operation on $X^*(T)$ is by mult).

Lemma 3.11 $X^*(\mathbb{G}_m) = \mathbb{Z}$

Proof: Consider $\theta: \mathbb{Z} \rightarrow X^*(\mathbb{G}_m)$
 $n \mapsto [t \mapsto t^n]^*$

Why $\mathcal{O}(n) := [t \mapsto t^n]^*$ is a hom of alg groups?

we need

$$\begin{array}{ccc} \mathbb{G}_m \times \mathbb{G}_m & \xrightarrow{m} & \mathbb{G}_m \\ \downarrow \mathcal{O}(n) \times \mathcal{O}(n) & \curvearrowright & \downarrow \mathcal{O}(n) \\ \mathbb{G}_m \times \mathbb{G}_m & \xrightarrow{m} & \mathbb{G}_m \end{array}$$

i.e. $m^* \circ \mathcal{O}(n)^* = (\mathcal{O}(n), \mathcal{O}(n))^* m^*$ $\mathcal{O}(n)^*: k[t, t^{-1}] \rightarrow k[t, t^{-1}]$
 $t \longmapsto t^n$

$$\begin{aligned} m^* \circ \mathcal{O}(n)^*(t) &= m^*(t^n) \\ &= (t \otimes t)^n \\ &= t^n \otimes t^n \\ &= (\mathcal{O}(n), \mathcal{O}(n))^* m^*(t). \end{aligned}$$

Next, $\mathcal{O}: \mathbb{Z} \rightarrow X^*(\mathbb{G}_m)$ is a hom of groups because
 $\mathcal{O}(a+b)(t) = t^{a+b} = t^a t^b = \mathcal{O}(a)(t) \cdot \mathcal{O}(b)(t).$

Injectivity: Clear.

Surjectivity: Let $\phi \in \text{Hom}(\mathbb{G}_m, \mathbb{G}_m)$. Write $\phi^*(t) \in k[t, t^{-1}]$
 and $\phi \neq 0$. as $\phi^*(t) = \sum_{i \in \mathbb{Z}} a_i t^i$, $i \in \mathbb{Z}$, $\text{lik } m$.

We have $m^*(\phi^*(t)) = \phi^*(t) \otimes \phi^*(t)$

$$\Rightarrow m^*\left(\sum a_i t^i\right) = \sum a_i t^i \otimes \sum a_j t^j$$

$$\Rightarrow \sum a_i t^i \otimes t^i = \sum a_i a_j t^i \otimes t^j$$

$\Rightarrow$ at most one a_i is non-zero, say a_n .

$$\Rightarrow a_n t^n \otimes t^n = a_n^2 t^n \otimes t^n$$

$$\Rightarrow a_n = 1.$$

$$\therefore \phi^*(t) = t^n, \text{ i.e., } \phi = \mathcal{O}(n).$$

• Linear representation of alg groups and tori:

- Let $\dim(V) < \infty$. Then $GL(V)$ is an affine alg group.
finite dim
- A linear rep of an alg group G on V is a hom of alg groups
 $\rho: G \longrightarrow GL(V)$.

• Action of an affine alg group G on a k -vect space V :

- $\forall$ k -alg R , a group action
 $\sigma_R: G(R) \times (V \otimes_k R) \longrightarrow V \otimes_k R$ such that
- $\sigma(g, -)$ is a hom of R -modules $\forall g$.
- functorial in R .

- Let $\rho: T \longrightarrow GL(V)$ be a finite dim lin rep of a torus.
Then T acts on V in the following way
$$\sigma_R: T(R) \times V \otimes_k R \longrightarrow V \otimes_k R$$
$$(g, v) \longmapsto \rho(g)(v)$$

This action gives a decomposition of V as:

Proposition 3.12: Any finite dimensional linear representation of a torus

$$\rho: T \longrightarrow GL(V)$$

gives a decomposition

$$V \cong \bigoplus_{\chi \in X^*(T)} V_\chi$$

where

$V_\chi := \{v \in V \mid t \cdot v = \chi(t)v \ \forall t \in T\}$ is called a **weight space**
and $\chi := \{\chi \mid V_\chi \neq 0\}$ is called a **weight of the action**.

Definition 3.13: G affine alg grp acting on a scheme X ,
 x a k -point of X .

i) The **orbit** $G \cdot x$ of x is the set-theoretic image of the morph

$$\sigma_x := \sigma(-, x) : G(k) \rightarrow X(k)$$

$$g \mapsto g \cdot x := \sigma(g, x)$$

ii) The stabilizer of x is the **fiber product**

$$\begin{array}{ccc} G_x := G \times_{X, \sigma_x} \text{Spec } k & \longrightarrow & \text{Spec } k \\ \downarrow & & \downarrow \sigma_x \\ G & \xrightarrow{\sigma_x} & X \end{array}$$

Remark: x a k -point means $x: \text{Spec } k \rightarrow X$ is closed emb.
 $\therefore G \times_{X, \sigma_x} \text{Spec } k \rightarrow G$ is base-change of closed embedding x and

hence is a closed embedding. Thus, G_x is a closed subscheme of G . Moreover, it is a subgroup since the fiber prod is in the category of affine alg groups.

Proposition 3.15: G affine alg group and $G \curvearrowright X$. Then:

(i) The orbits of closed points are locally closed subsets of X .

(ii) $\text{Bd}(G \cdot x) := \overline{G \cdot x} - G \cdot x = \text{union of orbits of smaller dim.}$

Defⁿ 3.16: An action of affine alg grp G on a scheme X is **closed** if all G -orbits in X are closed.

Exercise Sheet 3

Q.1. Consider the action of $GL_m \times GL_n$ on $Mat_{m \times n}$ by
 $(A, B) \circ M = AMB^{-1}$.

Classify

- ① All orbits of this action
- ② Open orbits
- ③ Closed orbits
- ④ Closure of orbits.

Ans: ① For a closed point $M \in Mat_{m \times n}$, we have.

$$\mathcal{O}_M = \{ N \in Mat_{m \times n} \mid N = AMB^{-1}, A \in GL_m, B \in GL_n \}$$

Now suppose $X \in Mat_{m \times n}$ and $\text{rank}(X) = r$. Then there exists $A \in GL_m, B \in GL_n$ s.t. $AXB^{-1} = \begin{bmatrix} I_r & \\ & 0 \end{bmatrix}$.

Therefore, $\mathcal{O}_M = \mathcal{O}_r := \{ N \in Mat_{m \times n} \mid \text{rk}(N) = \text{rk}(M) =: r \}$.

Thus, orbits are $\mathcal{O}_r$ where $0 \leq r \leq \min\{m, n\}$.

④ Closure of any orbit $\overline{\mathcal{O}_r} = \bigcup_{s \leq r} \mathcal{O}_s$:

Suppose $s < r$. Consider $M_t := \begin{bmatrix} I_s & & \\ & tI_{r-s} & \\ & & 0 \end{bmatrix} \in Mat_{m \times n}$

where $t \neq 0$. Then $M_t \in \mathcal{O}_r$. Now, for $t \rightarrow 0$, we see that $M_t \rightarrow \begin{bmatrix} I_s & \\ & 0 \end{bmatrix} \in \mathcal{O}_s$.

$\therefore \begin{bmatrix} I_s & \\ & 0 \end{bmatrix} \in \overline{\mathcal{O}_r}$ and hence $\mathcal{O}_s \subseteq \overline{\mathcal{O}_r}$ since we can reach any elt of $\mathcal{O}_s$ from $\begin{bmatrix} I_s & \\ & 0 \end{bmatrix}$.

$$\therefore \bigcup_{s \leq r} \mathcal{O}_s \subseteq \overline{\mathcal{O}_r}.$$

Finally, we claim that if $s > r$, then no element of $\mathcal{O}_s$ can be in $\overline{\mathcal{O}_r}$.

Suppose $N \in \mathcal{O}_s$, i.e., $\text{rk}(N) = s$ and $s > r$ and $M_k \rightarrow N$ where $M_k \in \mathcal{O}_r$

Now $M_k \in \mathcal{O}_r \Rightarrow$ All minors of order greater than r vanish

Since minors are polynomials in the entries of the matrix, each minor is continuous in the entries of the matrix.

$\therefore$ All minors of N of order greater than r vanish.

This contradicts $\text{rank}(N) = s > r$ because rank s is the largest integer such that some minor of N of order s does not vanish.

③ Closed orbits: Must satisfy $\overline{\mathcal{O}_r} = \mathcal{O}_r$

$$\Rightarrow \bigcup_{s \leq r} \mathcal{O}_s = \mathcal{O}_r$$

$$\Rightarrow r = 0$$

$\therefore$ Only closed orbit is $\mathcal{O}_0 = \{0_{m \times n}\}$.

② Open orbits: $\mathcal{O}_r$ open $\Leftrightarrow \mathcal{O}_r^c = \bigcup_{r \neq s} \mathcal{O}_s$ closed

We saw that for $\delta < r$, we can get a set $\{M_t\}$ of matrices in $\mathcal{O}_\delta$ having a limit point in $\mathcal{O}_\delta$.

Thus, if $0 \leq r < \min\{m, n\}$, then we can get a set in $\mathcal{O}_{r+1}$ whose limit point is in $\mathcal{O}_r$. But $\mathcal{O}_r \not\subseteq \mathcal{O}_r^c = \bigcup_{\delta \neq r} \mathcal{O}_\delta$

$\therefore \mathcal{O}_r^c$ is NOT closed if $0 \leq r < \min\{m, n\}$.

If $r = \min\{m, n\}$. Then $\mathcal{O}_r^c = \bigcup_{\delta \neq r} \mathcal{O}_\delta = \bigcup_{\delta \leq r-1} \mathcal{O}_\delta = \overline{\mathcal{O}_{r-1}} = \text{closed}.$

Thus, the unique open orbit is $\{M \mid \text{rank}(M) = \min\{m, n\}\}$, i.e., the orbit of matrices of maximal rank.

—————X—————